Supplementary Material

Lead-Time Simple Exponential Smoothing as a Configurable Framework for Adaptive Radiotherapy Monitoring

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Introduction

The following material provides mathematical details and exploratory analyses that extend the main manuscript while preserving its focus on lead-time simple exponential smoothing (LT-SES) as a configurable framework for adaptive radiotherapy monitoring. Section S1 develops the SES–ARIMA representation and the propagation of future innovations into the remaining-course aggregate. Section S2 describes the empirical residual-bootstrap construction and its fast Fourier transform (FFT) implementation. Section S3 presents a descriptive predictive comparison of SES with alternative forecasting models. Section S4 summarizes our institutional experience and practical considerations that motivated the LT-SES workflow, including dose-metric selection, selection of the smoothing parameter, and the use of probability-based prioritization for adaptive review. Displayed equations are numbered independently by section as (S1.x), (S2.x), and (S3.x); tables and figures use the corresponding section-based numbering.

Table of Contents

Introduction **S1**

S1 SES, ARIMA, and Lead-Time Innovation Propagation **S2**

S1.1 ARIMA Representation of the Fraction-Level Metric S2

S1.2 Derivation of the Remaining-Course Innovation Coefficients, Conditional Mean and Variance S3

S2 Empirical Residual Innovation Bootstrap Model **S5**

S2.1 Framework S5

S2.2 Propagation of Residual Innovations Through the SES State S6

S2.3 Discrete Bootstrap Probability Measures for Future Residual Innovations. S6

S2.4 Why FFT Avoids Bootstrap Combinatorics S7

S2.5 Computation of μ_{Q_h} using Convolution and the Discrete Fourier Transformation S7

S2.6 From the Residual Sum *CDF* to the Predictive *CDF* S8

S3 Descriptive Predictive Comparison of SES and Alternative Forecasting Models **S10**

Methods S10

Results S11

Parotid Examples S13

Interpretation and Limitations S15

S4 Practical Considerations and Institutional Experience **S16**

S4.1 Why We Pursued LT-SES S16

S4.2 A Practical Choice of Dose Metrics S16

S4.3 Choosing Alpha for Routine Monitoring S16

S4.4 From Triggers to Prioritization S17

S4.5 Uncertainty and the Value of a Simple Monitoring Layer S17

Supplementary Section S1: SES, ARIMA, and Lead-Time Innovation Propagation

S1.1. ARIMA Representation of the Fraction-Level Metric

The observed dose metric process $\{y_t\}$ has been modeled using SES with level ℓ_t and smoothing parameter α, and forecasts and prediction intervals have been constructed for the aggregate of the dose metric by summing the point forecasts. It is now shown that this forecasting scheme is the optimal predictor for a nonstationary ARIMA(0,1,1) dose metric process $\{y_t\}$ whose level is allowed to fluctuate and an ARIMA(0,2,1) stochastic process for the nonstationary aggregate $\{R_t\}$ of the dose metric process $\{y_t\}$.

Let $\{z_t\}$ be an arbitrary nonstationary stochastic process, B the backshift operator defined by the functional equation $Bz_t = z_{t-1}$, and $\{\varepsilon_t\}$ independent innovations that are identically distributed with mean 0 and variance σ^2. The linear stochastic process $\{\Delta_t\}$ given by

$$\Delta_t = \mu + \varepsilon_t + \psi_1\varepsilon_{t-1} + \psi_2\varepsilon_{t-2} + \cdots$$

is stationary if $\sum_{i=0}^{\infty} |\psi_i| < \infty$. A nonstationary stochastic process $\{z_t\}$ for which the *d-th* difference of the process is stationary can be represented by the following model:

$$\phi(B)\Delta_t = \theta(B)\varepsilon_t, \text{ where } \Delta_t \equiv (1-B)^d z_t \tag{S1.1}$$

And

$$\phi(B) \equiv 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p \tag{S1.2}$$

$$\theta(B) \equiv 1 - \theta_1 B - \theta_2 B^2 - \cdots - \theta_q B^q \tag{S1.3}$$

In practice, d is usually 0, 1, or at most 2, where $d = 0$ corresponds to the stationary behavior of the stochastic process. This class of processes are called *autoregressive integrated moving average* (ARIMA) processes of order (p, d, q). The reason for inclusion of the word 'integrated' in ARIMA is as follows. Inverting $\Delta_t = (1-B)^d z_t$ yields $z_t = S^d\Delta_t$, where $S = (1-B)^{-1}$ is the *summation operator* defined by

$$S\Delta_t = \sum_{j=0}^{\infty} \Delta_{t-j} = \Delta_t + \Delta_{t-1} + \Delta_{t-2} + \cdots$$

Thus, a general ARIMA process may be generated by summing or 'integrating' the process $\{\Delta_t\}$ d times.

In what follows y_t denotes the dose metric at fraction t, for which the difference in the delivered dose metric to the previous fraction is given by

$$\Delta_t = y_t - y_{t-1} = (1-B)\, y_t. \tag{S1.4}$$

Because the delivered dose metric is allowed to drift over the course of treatment, the dose metric process $\{y_t\}$ is treated as nonstationary but it is assumed that the process of the differences $\{\Delta_t\}$ comes from a stationary process. Then $\{y_t\}$ can be modeled as an ARIMA(0,1,1) process which by equations (S1.1), (S1.2), and (S1.4) is given by:

$$(1-B)\, y_t = (1-\theta_1 B)\ \varepsilon_t \tag{S1.5}$$

where $|\theta_1| < 1$. Inverting, (S1.5) yields:

$$y_t = (1-B)^{-1}\,(1-\theta_1 B)\,\varepsilon_t, \tag{S1.6}$$

Since $R_t = (1-B)^{-1} y_t$, we find that

$$R_t = (1-B)^{-2}\,(1-\theta_1 B)\,\varepsilon_t, \tag{S1.7}$$

Hence, with the assumptions we have made for the process $\{y_t\}$ this shows that the nonstationary aggregate $\{R_t\}$ is an ARIMA(0,2,1) process. It can be shown that the process in equation (S1.6) can be written as the weighted average of previous values of the process:

$$y_t = \widehat{y}_{t|t-1}(\alpha) + \varepsilon_t \tag{S1.8}$$

$$\widehat{y}_{t|t-1}(\alpha) = \alpha \sum_{j=1}^{\infty} (1-\alpha)^{j-1} y_{t-j} \tag{S1.9}$$

and $\alpha = 1 - \theta_1$, hence $0 < \alpha < 2$ (cf. Ref. [1]). As discussed in Box et al. [1] $\widehat{y}_{t|t-1}(\alpha)$ is the optimal linear one-step-ahead predictor, also referred to as the level, of the process $\{y_t\}$, i.e. what is known about the future value of this process. Using equation (S1.9) a recursive relationship for this optimal one-step-ahead predictor of y_t is now derived, which shows that $\widehat{y}_{t|t-1}(\alpha)$ is indeed our simple SES model

$$\begin{aligned}
\widehat{y}_{t+1|t}(\alpha) &= \alpha \sum_{j=1}^{\infty} (1-\alpha)^{j-1} y_{t-(j-1)} = \alpha \sum_{k=0}^{\infty} (1-\alpha)^{k} y_{t-k} \\
\widehat{y}_{t+1|t}(\alpha) &= \alpha y_t + \alpha \sum_{k=1}^{\infty} (1-\alpha)^{k} y_{t-k} = \alpha y_t + (1-\alpha)\alpha \sum_{k=1}^{\infty} (1-\alpha)^{k-1} y_{t-k} \\
\widehat{y}_{t+1|t}(\alpha) &= \widehat{y}_{t|t-1}(\alpha) + \alpha[y_t - \widehat{y}_{t|t-1}(\alpha)].
\end{aligned} \tag{S1.10}$$

Setting $\ell_t = \widehat{y}_{t+1|t}(\alpha)$ we find

$$\ell_t = \ell_{t-1} + \alpha\,(y_t - \ell_{t-1}) \tag{S1.11}$$

This recursive relationship is easily derived from the SES state model shown in equation (3). Thus, the SES model for the per-fraction dose metric is the optimal linear one-step-ahead forecast under an ARIMA(0,1,1) model for the nonstationary dose metric process $\{y_t\}$ and an ARIMA(0,2,1) process for its nonstationary aggregate $\{R_t\}$.

In the traditional SES setting, one restricts $0 < \alpha \leq 1$, for which the SES level is an exponentially weighted moving average with nonnegative weights that sum to one. Although the SES–ARIMA(0,1,1) equivalence implies a broader admissible region $0 < \alpha < 2$ under invertibility, an $\alpha > 1$ yields alternating-sign weights, and therefore does not retain the standard smoothing interpretation; for interpretability in ART one therefore, restricts α to the half open interval $(0, 1]$. Also note that allowing the value $\alpha = 1$, shows that LOCF is one particular case of the SES strategy for adaptive monitoring being discussed in this work (see equation (S1.11) since the level ℓ_t for any $t \in \{1, \cdots, N\}$ is then simply equal to the observed value y_t.

S1.2. Derivation of the Remaining-Course Innovation Coefficients, Conditional Mean and Variance

Starting from the state equation at fraction t,

$$\begin{aligned}
y_{t+1} &= \ell_t + \varepsilon_{t+1} \\
\ell_{t+1} &= \ell_t + \alpha\varepsilon_{t+1},
\end{aligned}$$

One straightforwardly finds by backwards substitution that

$$y_{t+2} = \ell_t + \alpha\varepsilon_{t+1} + \varepsilon_{t+2},$$

and, more generally for carrying out the indicated backwards substitution $j - 1$ times,

$$y_{t+j} = \ell_t + \alpha \sum_{r=1}^{j-1} \varepsilon_{t+r} + \varepsilon_{t+j}.$$

Forming the aggregate of future values by summing from $j = 1$ to h yields

$$R_{t+h} = h\ell_t + \sum_{j=1}^{h} \left(\varepsilon_{t+j} + \alpha \sum_{r=1}^{j-1} \varepsilon_{t+r} \right).$$

To make it clear that the random variable R_{t+h} is based on SES and therefore depends on the level ℓ_t and the innovations, we will use the notation $R_{t+h|t}$ hereafter. Note, that a future innovation ε_{t+r} appears once directly and in each subsequent forecasted observation through the state update. If one writes out the sum over h in lower diagonal matrix form and then sums up each of the resulting columns one finds the total coefficient for a given ε_{t+r} to be:

$$c_r = 1 + \alpha(h - r), \qquad r = 1, \ldots, h. \tag{S1.12}$$

Hence,

$$R_{t+h|t} = h\ell_t + \sum_{r=1}^{h} c_r \varepsilon_{t+r}. \tag{S1.13}$$

The sum in equation (S1.13) can be rewritten as follows

$$\sum_{r=1}^{h} c_r \varepsilon_{t+r} = \sum_{k=0}^{h-1} c_{h-k} \varepsilon_{t+h-k},$$

and hence, equation (S1.13) becomes

$$R_{t+h|t} = h\ell_t + \sum_{k=0}^{h-1} c_{h-k} \varepsilon_{t+h-k}. \tag{S1.14}$$

Using equation (S1.14) one can now easily compute conditional expectation of R_{t+h} as follows;

$$\begin{aligned}
\widehat{R}_{t+h|t} &\equiv E(R_{t+h|t} \mid \mathcal{F}_t) \\
&= E(h\ell_t + \sum_{k=0}^{h-1} c_{h-k} \varepsilon_{t+h-k} \mid \mathcal{F}_t) \\
&= h\ell_t + \sum_{k=0}^{h-1} c_{h-k} E(\varepsilon_{t+h-k} \mid \mathcal{F}_t) \\
&= h\ell_t,
\end{aligned}$$

where we have used following well known fact about the conditional expectation namely, $E(a + bX \mid Y) = a + bE(X \mid Y)$ and that $E(\varepsilon_{t+h-k} \mid \mathcal{F}_t) = 0$ for all k by assumption. For independent innovations as assumed here with common variance σ^2 one finds for the conditional variance

$$\begin{aligned}
V\left(R_{t+h|t} \mid \mathcal{F}_t\right) &= V\left(h\ell_t + \sum_{k=0}^{h-1} c_{h-k} \varepsilon_{t+h-k} \mid \mathcal{F}_t\right) \\
&= V\left(\sum_{k=0}^{h-1} c_{h-k} \varepsilon_{t+h-k} \mid \mathcal{F}_t\right) \\
&= \sum_{k=0}^{h-1} c_{h-k}^2 V\left(\varepsilon_{t+h-k} \mid \mathcal{F}_t\right) \\
&= \sum_{k=0}^{h-1} c_{h-k}^2 V\left(\varepsilon_{t+h-k}\right) \\
&= \sigma^2 \sum_{k=0}^{h-1} c_{h-k}^2,
\end{aligned}$$

where the following well known facts about the conditional variance of independent random variables have been used in the second, third, and fourth equation, above namely, $V(a + bX \mid Z) = V(bX \mid Z)$, $V(aX + bY \mid Z) = a^2V(X \mid Z) + b^2V(Y \mid Z)$, and $V(X \mid Y) = V(X)$. The last equation follows from the fact that innovations are identically distributed with common variance σ^2. Using equation (S1.12) yields the following result.

$$\sum_{k=0}^{h} c_{h-k}^2 = \sum_{k=0}^{h-1} (1 + \alpha k)^2 = h + 2\alpha \sum_{k=0}^{h-1} k + \alpha^2 \sum_{k=0}^{h-1} k^2,$$

which after summing over k and simplifying yields the closed-form expression for $\Omega(h, \alpha)$ given in equation (9) in the main text.

Supplementary Section S2: Empirical Residual Innovation Bootstrap Model

The analytic Gaussian formulation provides a compact, closed-form predictive distribution, but its shape is determined by the assumption of having independent, normally distributed innovations with mean zero and common standard σ^2. This appendix develops a model-based empirical residual bootstrap in which the observed post-initialization innovations are centered, i.e. have mean zero, and are treated as an empirical distribution for future innovations. These values are propagated through the same SES state dynamics and lead-time coefficients as used in the analytic formulation. The method therefore preserves the underlying forecast structure while allowing the predictive distribution to reflect features of the observed residuals, including skewness, tail asymmetry, and discreteness. Table S2.1 summarizes the principal differences between the two approaches.

Table S2.1: Comparison of the analytic and empirical predictive-distribution constructions. FFT: fast Fourier transform; SES: simple exponential smoothing.

Feature	Analytic Gaussian formulation	Empirical residual-bootstrap formulation
SES state dynamics	Retained	Retained
Innovation distribution	Gaussian with estimated variance	Empirical centered residual distribution
Predictive output	Closed-form normal distribution	Discrete predictive probability distribution
Computation	Direct formula	Grid construction and FFT convolution
Principal advantage	Maximum transparency and minimal computation	Can retain observed skewness, tail asymmetry, or discreteness
Principal limitation	Sensitive to non-Gaussian and heteroscedastic innovations	Limited by residual sample size and still assumes future residuals are represented by the empirical distribution of past residuals

S2.1. Framework

Denote by

$$\mathcal{E}_t = \{\varepsilon_1, \cdots, \varepsilon_t\}$$

the set of observed one-step-ahead residual innovations from an SES forecasting model. These observed residual innovations are treated as an empirical discrete sample for the future residual innovation distribution. If no weighting of the innovations is assumed, i.e. each innovation has equal probability of occurring, then the empirical CDF can be written as:

$$\widehat{F}_{\varepsilon}(x) = \frac{1}{t}\sum_{i=1}^{t} H(x - \varepsilon_i),$$

where $H(x)$ denotes the Heaviside step function, which is defined as follows

$$H(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0. \end{cases}$$

In what follows, we choose to work with the centered residual innovations, i.e residual innovations that are centered around their mean. To this end set

$$\widetilde{\varepsilon}_i = \varepsilon_i - \overline{\varepsilon},$$

where

$$\overline{\varepsilon} = \sum_{i=1}^{t} \varepsilon_i$$

so that future residual innovations are bootstrapped around a mean of zero, rather than around $\overline{\varepsilon}$. The centered CDF then becomes:

$$\widehat{F}_{\widetilde{\varepsilon}}(x) = \frac{1}{t}\sum_{i=1}^{t} H(x - \widetilde{\varepsilon}_i).$$

S2.2. Propagation of Residual Innovations Through the SES State

When doing a multi-time step ahead prediction a future residual innovation can influence more than the single future observation for which it has been sampled. Because the SES state is updated after each future residual innovation, an earlier residual can also affect later forecasts. The cumulative residual contribution to the future total can therefore be written as a weighted residual innovation aggregate:

$$Q_h = \sum_{k=1}^{h} c_j \mathrm{E}_{t+j},$$

where h denotes the number of future time steps, E_{t+j} denotes the bootstrapped residual random variable at future time step j, c_j denotes the SES propagation coefficient for the residual innovation from j to h, and Q_h is the random variable representing the total future residual innovation contribution after SES propagation. Note that Q_h is the weighted sum of independent identically distributed random variables.

The coefficients are determined by the SES update equation employed and the selected smoothing and damping parameters. For the update equation, used in this work the c_j are given by $c_j = 1 + \alpha(h - j)$ (see Equation (S1.12) above). The reader is referred to Table 6.6 in Hyndman et al. [2] for the update equations of the most commonly used state space models having homoscedastic innovations.

Observe that the last future residual innovation contributes only directly with coefficient 1, while earlier residual innovations have larger effective coefficients because they also perturb subsequent SES states. We now show how one can estimate the distribution of Q_h using the convolution theorem and Discrete Fourier Transformation (DFT). However, before this can be done, one first needs to define a discrete probability measure for the independent and identically distributed random variables that make up Q_h.

S2.3. Discrete Bootstrap Probability Measures for Future Residual Innovations.

Denote by $(\mathcal{X}_t, \Omega)$ the measurable space one is working with. For each future time step j, define the scaled innovation random variable with compact support

$$X_j = c_j \mathrm{E}_{t+j}, \qquad j = 1, \cdots, h.$$

Under the empirical bootstrap described above, the possible realizations for each X_j are

$$x_{i,j} = c_j \widetilde{\varepsilon}_i, \qquad i = 1, \cdots, t$$

If the residual innovations are unweighted, each historical residual innovation contributes with equal probability, namely,

$$p_{i,j} = \frac{1}{t},$$

and one can define the following discrete probability measure for the X_j

$$\mu_j(A) = \frac{1}{t}\sum_{i=1}^{t} \delta_{x_{i,j}}(A), \quad j = 1, \cdots, h$$

where δ_x is the Dirac measure, which is defined as follows. For a fixed point $x \in \mathcal{X}_t$, the Dirac measure is defined for any set $A \in \Omega$ as

$$\delta_x(A) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A. \end{cases}$$

Since DFT operates on arrays, the values of the scaled innovation random variables X_j need to be placed on a common grid. Let $q < 1$ be the quantization step width, then one can define a common grid by rounding to the closest integer

$$k_{i,j} = \lfloor q x_{i,j} \rceil, \qquad i = 1, \cdots, t$$

where $\lfloor \cdot \rceil$ denotes the operation to round to the closest integer. The step specific probability measure associated with this grid definition is then given by

$$\mu_j(A) = \frac{1}{t} \sum_{i=1}^{t} \delta_{k_{i,j}}(A), \qquad j = 1, \cdots, h.$$

S2.4. Why FFT Avoids Bootstrap Combinatorics

A direct residual-bootstrap enumeration would consider all possible paths

$$(\mathrm{E}_{t+1},\ \mathrm{E}_{t+2}, \cdots, \mathrm{E}_{t+h}\ ).$$

With t historical residual innovations and h future time steps, a complete enumeration of paths requires enumeration of t^h distinct paths. Each path produces one specific realization of Q_h, namely,

$$q_h^{(p)} = \sum_{j=1}^{h} c_{h-j} \widetilde{\varepsilon}_{p_j}, \qquad \{p_1, \cdots, p_h\} \in \{1,\ \cdots, t\}^h$$

where $\{p_1, \cdots, p_h\}$ denotes one of the t^h possible permutations with replacement generated from the set $\{1,\ \cdots, t\}$. This enumeration is combinatorial. Complete enumeration quickly becomes impractical as h and t grow beyond small numbers: for 10 past time steps and a prediction horizon of 5, one would already have to enumerate 10^5 possible paths. Monte Carlo bootstrap avoids complete enumeration by using random sampling of paths; however, this introduces random sampling error, especially in the predictive tails of the estimated bootstrap distribution [3].

The DFT approach instead computes the discrete bootstrap probability measure associated with Q_h by convolution of the probability measures associated with each of the $X_j,\ j = 1, \cdots, h$.

S2.5. Computation of μ_{Q_h} using Convolution and the Discrete Fourier Transformation

Since Q_h is the sum of independent and identically distributed random variables assumed to have a common compact support,

$$Q_h = \sum_{j=1}^{h} X_j,$$

it can be bootstrapped using convolution and DFT [4]. The probability measure for Q_h can be found by convolution of the step specific probability measures

$$\mu_{Q_h} = \mu_1 * \mu_2 * \cdots * \mu_h$$

Applying DFT recursively to this expression and using the convolution theorem yields the following result.

$$\mathcal{F}\left(\mu_{Q_h}\right) = \prod_{j=1}^{h} \mathcal{F}\left(\mu_j\right),$$

and inversion yields

$$\mu_{Q_h} = \mathcal{F}^{-1}\left(\prod_{j=1}^{h} \mathcal{F}\left(\mu_j\right)\right)$$

Here DFT is used neither to estimate a residual spectrum nor to infer periodic structure, but to convert the operation of convolution to multiplication in the frequency domain and to transform the resulting product back to the time domain to compute the bootstrap probability measure for Q_h.

S2.6. From the Residual Sum *CDF* to the Predictive *CDF*

Once the probability measure for Q_h has been computed, it can be shifted and scaled to obtain the predictive discrete probability measure for the final quantity of interest. If D_t denotes the deterministic forecast aggregate and N the total number of fractions, then

$$\overline{Y} = g(Q_h) = \frac{D_t + Q_h}{N}.$$

Since g is one to one, the CDF for $\overline{Y}$ can be determined from the CDF of $Q_h$as follows:

$$F_{\overline{Y}}(y) = P\left(\overline{Y} \leq y\right) = P\left(Q_h \leq g^{-1}(y)\right) = P\left(Q_h \leq Ny - D_t\right) = F_{Q_h}\left(Ny - D_t\right).$$

Thus, the DFT convolved discrete residual bootstrap probability measure directly generates a predictive discrete probability measure for the final dose metric. The method is nonparametric with respect to the distribution of the residual innovations because the shape of the discrete $\mu_{Q_h}$is inherited from the empirical discrete sample of the residual innovations rather than imposed by a Gaussian probability measure.

S2.6.1 Predictive Performance Comparison to the Gaussian distribution

Predictive performance of the centered empirical residual-bootstrap and analytic Gaussian distributions was compared using the continuous ranked probability score (CRPS), which evaluates the complete predictive distribution against the realized value, with lower scores indicating better distributional performance [5]. For each sequence and forecast horizon, the relative paired difference was calculated as $\left(CRPS_{bootstrap} - CRPS_{Gaussian}\right) / CRPS_{Gaussian}$, so negative values favor the empirical FFT distribution and positive values favor the Gaussian distribution.

For visualization, cohort identity was displayed consistently across the supplementary comparisons: the total cohort is shown in dark gray, head and neck in blue, and parotid in orange. Reference lines were neutral, so the sign of a paired difference was conveyed by its position relative to zero rather than by color.

At $\alpha = 0.5$, the median relative differences were negative in 10 of the 12 cohort–horizon comparisons (Figure S2.1). For the total cohort, the medians were +0.02‰, −0.72‰, −1.41‰, and −1.98‰ at $h = 5$, 10, 15, and 20, respectively. The corresponding medians for the head-and-neck cohort were +0.19‰, −0.99‰, −1.87‰, and −2.45‰. While medians for the Parotid cohort were −0.68‰, −1.53‰, −1.78‰, and −0.73‰.

The median differences were therefore close to zero at $h = 5$ for the total and head-and-neck cohorts and modestly favored the empirical FFT distribution at longer horizons. The parotid medians were negative at all four horizons. The distributions nevertheless extended across zero in the displayed comparisons, indicating substantial overlap in per-series performance. The relative differences are small indicating the Gaussian approximation is suitable for LT-SES.

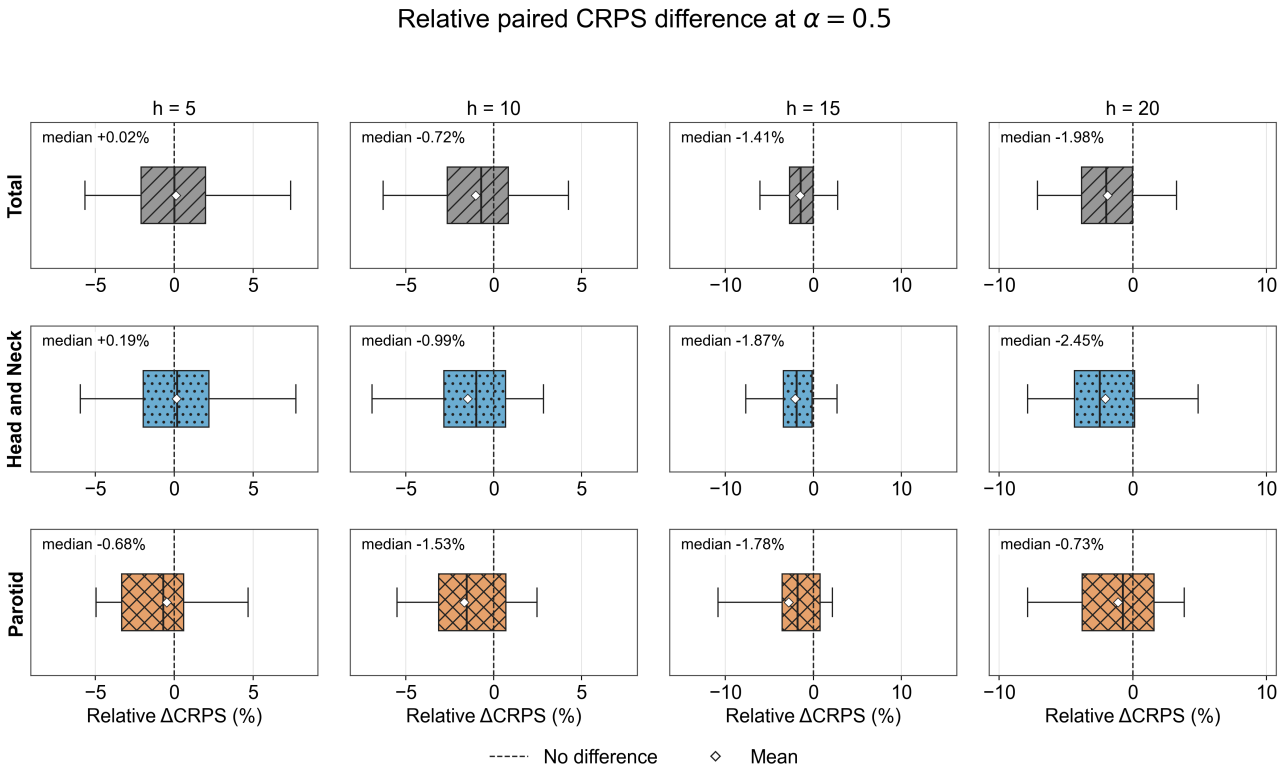


Figure S2.1: Relative paired continuous ranked probability score (CRPS) differences at $\alpha = 0.5$ for the total cohort ($n = 340$; top), head-and-neck cohort ($n = 181$; middle), and parotid cohort ($n = 44$; bottom), comparing the analytic Gaussian and centered empirical FFT-bootstrap predictive distributions. Positive values favor the Gaussian distribution and negative values favor the empirical FFT distribution. Boxes show interquartile ranges, whiskers show the 5th–95th percentiles, and diamonds show arithmetic means.

Supplementary Section S3: Descriptive Predictive Comparison of SES and Alternative Forecasting Models

Methods

SES using three fixed α-values (1.00, 0.50, and 0.25) was compared with an exchangeable Bayesian reference, series specific SES, Holt's linear method, and damped Holt's method. Forecasts used $h = 5$, 10, 15, or 20 fractions remaining and only data through $n = N - h$. The target was the realized final arithmetic mean,

$$\bar{y}_N = \frac{1}{N}\sum_{t=1}^{N} y_t. \tag{S3.1}$$

Each organ-at-risk structure time series was analyzed separately. Results are shown for the full cohort and the nested head-and-neck and parotid groups.

The Bayesian model assumed exchangeable normal observations with unknown mean and variance and prior $\pi(\mu, \sigma^2) \propto 1/\sigma^2$. Its predictive distribution for the final mean was

$$\overline{Y}_N \mid y_{1:n} \sim t_{n-1}\left(\bar{y}_n,\ s_n\sqrt{\frac{h}{n(n+h)}}\right), \tag{S3.2}$$

where s_n is the sample standard deviation of the observed prefix and the second argument denotes the Student-t scale. Because the Bayesian forecast depends only on the prefix mean and standard deviation, reordering observations does not change it. The exponential-smoothing models retain time order: SES tracks a local level, Holt's linear method an undamped trend, and damped Holt's method a damped trend. These correspond to ETS(A,N,N), ETS(A,A,N), and ETS(A,A_d,N), respectively [6–8].

Visual encodings were held consistent across the supplementary comparisons. Cohort summaries use dark gray for the full cohort, blue for head and neck, and vermillion for parotid, with distinct line, marker, or hatch patterns where applicable. Model colors are held constant across panels. In the HN3 and HN15 examples, SES with $\alpha = 0.25$ is shown as a blue solid curve and the Bayesian reference as a dark-gray dashed curve, with matching shaded prediction intervals and neutral threshold lines. Printed labels and distinct line or marker patterns provide additional identification.

Table S3.1: Fitted quantities and the corresponding K used in the exponential-smoothing innovation-variance estimate.

Model	Fitted from the observed prefix	Plain-language role	K
Bayesian reference	μ and σ^2 integrated over their posterior	Common mean and between-fraction spread; all observations have equal weight	N/A
SES, $\alpha = 1.00$	Initial level ℓ_0	Starting level; the current level then follows the latest observation	1
SES, $\alpha = 0.50$	Initial level ℓ_0	Starting level with moderate response to new observations	1
SES, $\alpha = 0.25$	Initial level ℓ_0	Starting level with a smoother, slower response	1
Series-specific SES	ℓ_0 and α	Starting level and fitted responsiveness	2
Holt's linear method	ℓ_0, initial trend b_0, α, and β	Level plus an undamped trend; β controls trend updating	4
Damped Holt's method	ℓ_0, b_0, α, β, and damping ϕ	Level plus a trend whose future effect decreases when $\phi < 1$	5

Note. K is the number of parameters fitted before estimating the innovation variance; the innovation variance itself is not included. The Bayesian model uses its Student-t posterior predictive distribution and does not use this correction.

At each forecast origin, all free quantities were estimated from the available prefix by minimizing the sum of squared one-step ahead innovations (SSE). For fixed-α SES, only ℓ_0 was fitted. For series-specific SES, ℓ_0 was re-estimated for each candidate α before selecting the pair with the lowest error. Holt's linear and damped Holt methods were fitted in the same manner. The plug-in innovation variance was

$$\widehat{\sigma}_n^2 = \frac{\mathrm{SSE}_n}{n - \mathrm{K}}. \tag{S3.3}$$

The fitted state and covariance were propagated over the remaining h fractions, combined with the observed prefix sum, and divided by N to obtain the final-mean predictive distribution. Predictive performance was characterized using CRPS, for which lower values are better [5]. CRPS values were normalized after forecasting by the realized full-course mean.

A separate cohort profile characterized one-step SES error. Each complete sequence was normalized by its full-course mean, ℓ_0 was re-estimated for every candidate α, and squared innovations were pooled within each cohort. This profile was descriptive; the predictive comparison used the same fixed value, $\alpha = 0.25$, for all cohorts.

Results

The profiled one-step mean squared error (MSE) attained its minimum at $\alpha = 0.22$ for the full cohort, 0.25 for the head and neck cohort, and 0.27 for the parotid cohort (Figure S3.1). As can be seen from Figure S3.1 the curves were shallow near their minima, and the common value $\alpha = 0.25$ was close to all three. These values describe this particular dataset and are not clinical prescriptions.

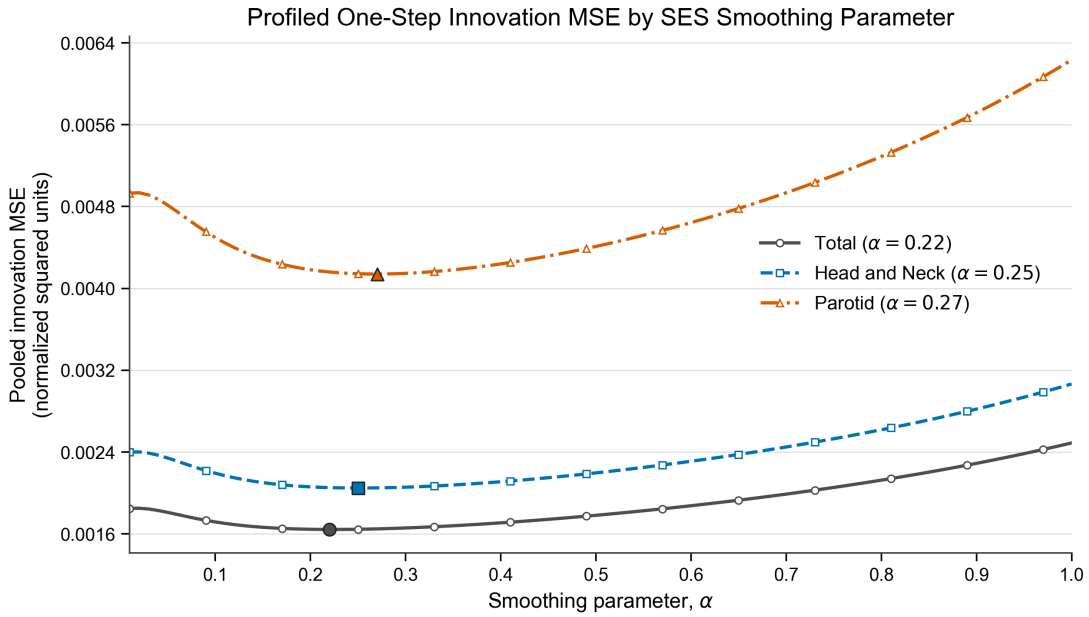


Figure S3.1: Profiled one-step innovation MSE by SES smoothing parameter. The initial level was re-estimated for every candidate α. The full cohort is shown in dark gray with circles and a solid line, the head and neck cohort in blue with squares and a dashed line, and the parotid cohort in vermillion with triangles and a dash-dot line. The later predictive comparison used fixed $\alpha = 0.25$ in all cohorts.

Fixed SES with $\alpha = 0.25$ had the lowest arithmetic-mean CRPS in all 12 cohort–horizon panels (Figure S3.2). The Bayesian model and SES with $\alpha = 0.50$ were generally the closest alternatives, with substantial overlap among distributions. Series-specific fitting of α did not provide a consistent improvement. Holt's linear method, which includes an undamped trend, produced the largest long-horizon errors; damped Holt reduced, but did not remove, this behavior by damping the trend.

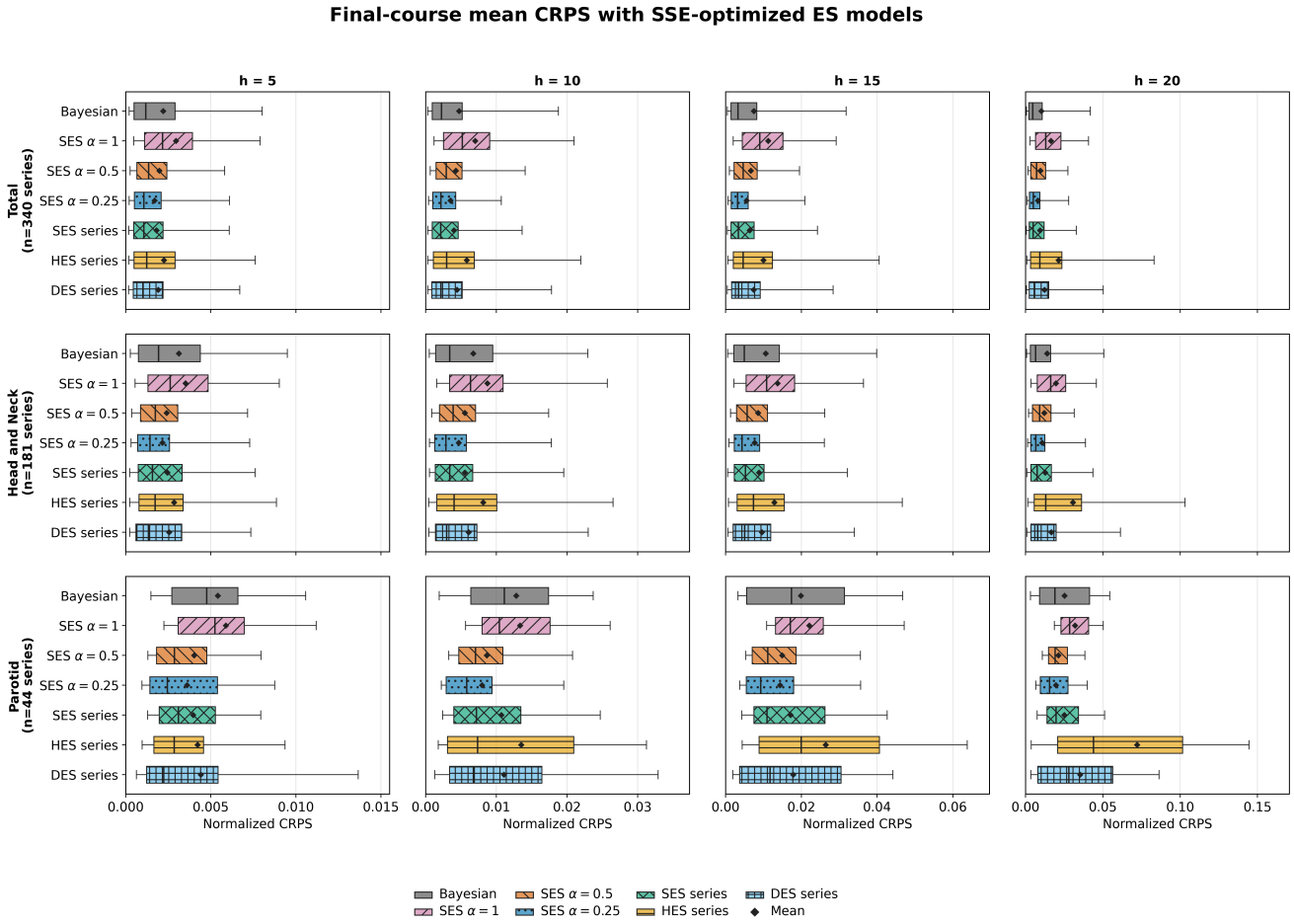


Figure S3.2: Per-series normalized CRPS at horizons of 5, 10, 15, and 20 remaining fractions. Plot labels HES and DES correspond to Holt's linear method and damped Holt's method, respectively. Model colors are held constant across panels: Bayesian is dark gray; SES with $\alpha = 1.0$, 0.50, and 0.25 is reddish purple, vermillion, and blue, respectively; series-specific SES is bluish green; HES is orange; and DES is sky blue. Boxes retain neutral outlines and printed model labels. Boxes show interquartile ranges, center lines show medians, whiskers show the 5th and 95th percentiles, and diamonds show arithmetic means. Lower CRPS is better.

The within-series comparison showed the same general pattern (Figure S3.3). Differences were calculated as Bayesian CRPS minus SES CRPS, so positive values favor SES with $\alpha = 0.25$. Medians and arithmetic means were positive in every panel, but the distributions also extended below zero. Thus, SES performed better for the typical series, but not for every series.

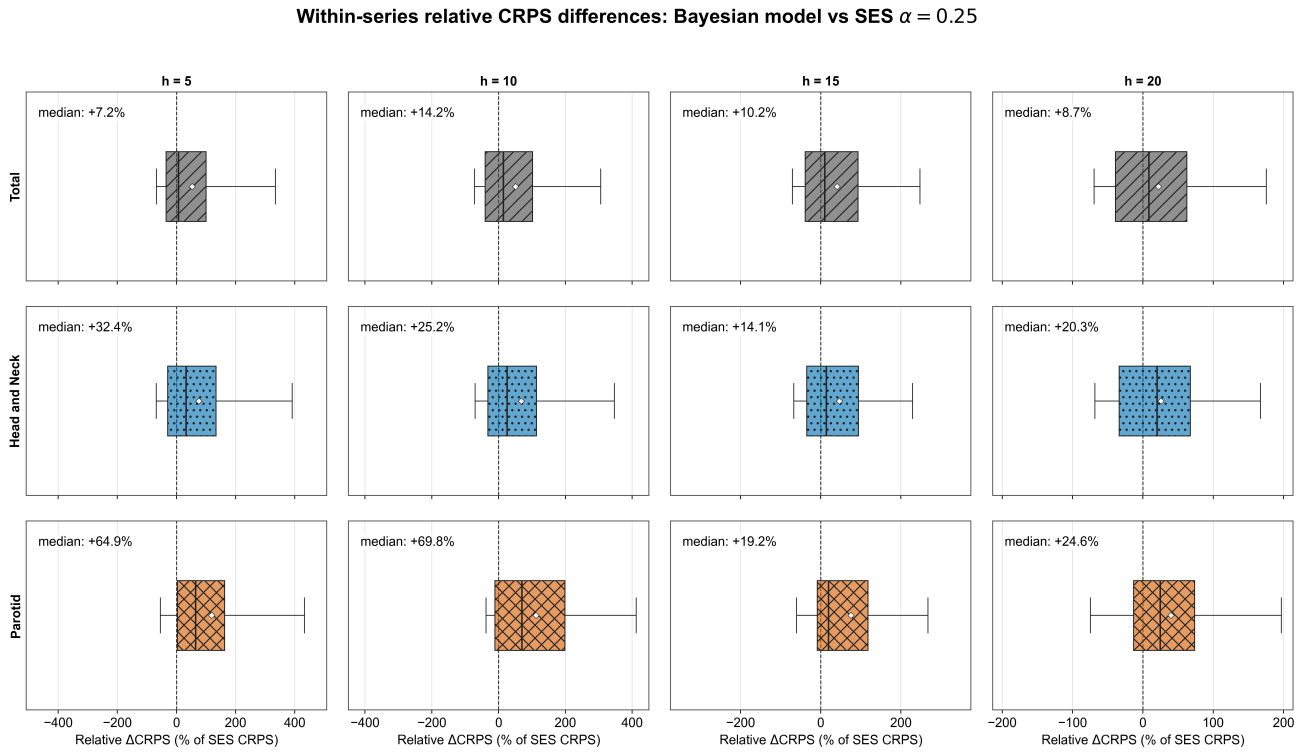


Figure S3.3: Within-series normalized CRPS difference between the Bayesian model and SES with fixed $\alpha = 0.25$. The full, head-and-neck, and parotid cohorts use the common dark-gray, blue, and vermillion mapping, respectively, with a neutral zero-reference line. Positive values favor SES and negative values favor the Bayesian model by position relative to zero. These are descriptive structure-level comparisons, not patient-level confidence intervals.

Parotid Examples

The examples illustrate why time order may have an impact on patient management during ART. The Bayesian model gives equal weight to all observed fractions, whereas SES places the highest weight to the most recent observations. In HN3, which has marked day-to-day variation, SES responds sooner to recent low- and high-dose clusters, while the Bayesian forecast is more stable (see Figure S3.4). This responsiveness changes the projected final mean and threshold probabilities earlier, but it can also reflect short-lived variation.

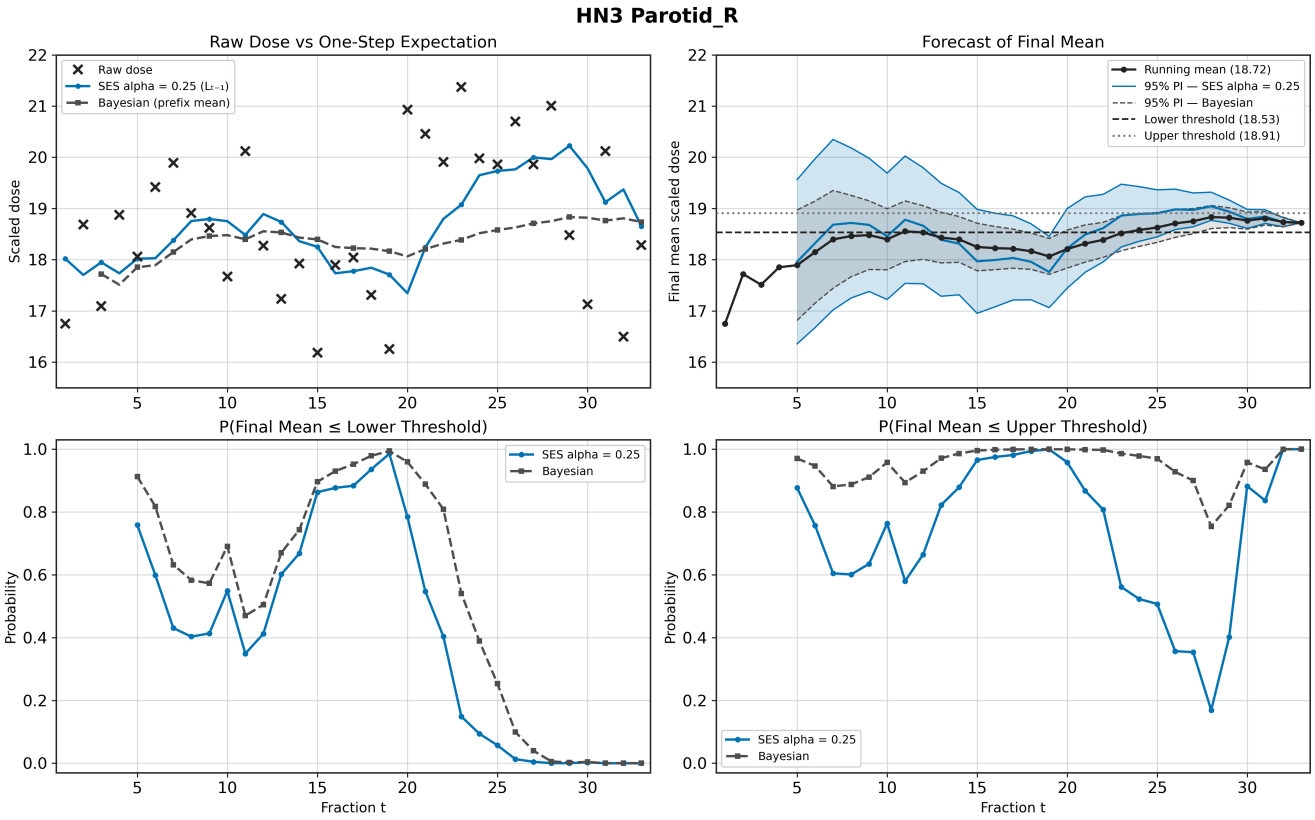


Figure S3.4: HN3 right-parotid example. SES with $\alpha = 0.25$ responds sooner to recent clusters of observations, whereas the exchangeable Bayesian model changes through the cumulative prefix mean and variance. SES is shown as a blue solid curve and the Bayesian reference as a dark-gray dashed curve, with matching blue and gray prediction-interval fills; threshold lines are neutral and distinguished by dash pattern. Thresholds are illustrative and are not clinical action levels.

In example HN15, a sustained increase begins near the middle of the course (see Figure S3.5). SES moves toward the new local level sooner. Its lower-threshold probability falls sharply around fractions 20 – 22, and its upper-threshold probability temporarily decreases, while the Bayesian upper-threshold probability remains near one. The forecasts converge only after more post-change observations accumulate.

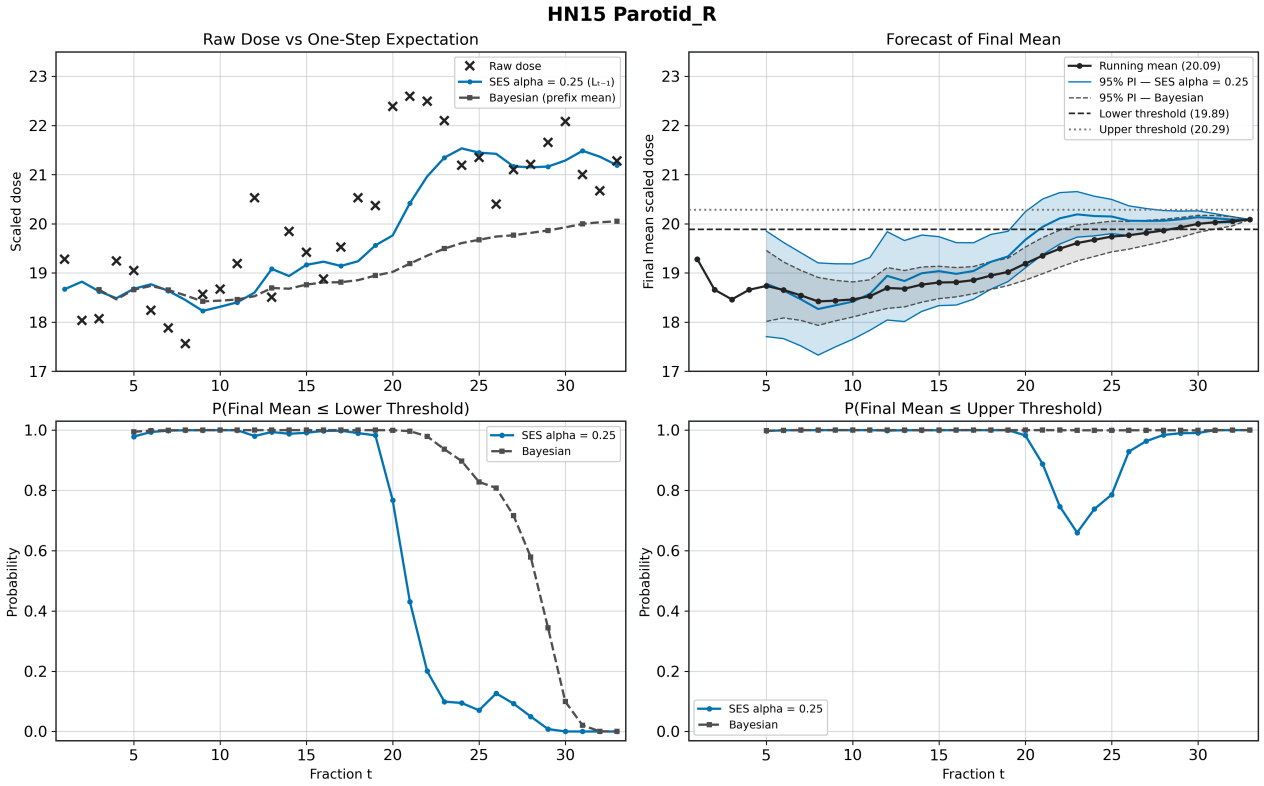


Figure S3.5: HN15 right-parotid example with a sustained midcourse increase. SES with $\alpha = 0.25$ adapts sooner to the new local level; the Bayesian prefix mean changes more slowly because observation order does not affect the model. SES is shown as a blue solid curve and the Bayesian reference as a dark-gray dashed curve, with matching blue and gray prediction-interval fills and neutral threshold lines.

Interpretation and Limitations

Overall, the results support the use of a relatively simple local-level model for these sparse fraction-level time series. These findings are specific to the measured dataset evaluated here and should not be interpreted as establishing a universally optimal model or value of α. The fixed value of $\alpha = 0.25$ was a useful retrospective reference; clinical selection of α should still reflect the desired balance between responsiveness and stability. This selection configures the monitoring response and does not calibrate the resulting threshold probabilities. For ranking-only applications evaluation should focus on ordering and comparability; absolute probability interpretation requires a separate probability-calibration assessment. The cohorts were nested, several structures came from the same patient, and only one forecast origin was tested for each time series and prediction horizon. The results therefore describe time series-level performance rather than patient-level inference. Residual autocorrelation, changing variance, nonlinear drift, and abrupt regime changes were not explicitly modeled.

Supplementary Section S4: Practical Considerations and Institutional Experience

The preceding sections describe the technical basis and exploratory evaluation of LT-SES. This section provides a different perspective. It summarizes the practical experience that motivated the framework and the choices that were made by us while developing it for adaptive radiotherapy monitoring. These observations reflect our institutional experience and should not be interpreted as validated clinical recommendations.

S4.1. Why We Pursued LT-SES

Our work began by exploring alternatives to the last-observation-carried-forward (LOCF) approach used by the commercial software available in our clinical workflow. LOCF provided a simple projection of the most recent fraction-level dose metric across the remaining treatment course, but we wanted an approach that could incorporate information from the preceding realized values in the fraction time series rather than relying entirely on the latest observation. This led us to explore simple exponential smoothing and subsequently to use SES for process control of patient treatments, as described in our earlier work [9]

As we gained experience with SES, we wanted to provide more information than either process-control output or a simple scalar difference between an observed or projected dose metric and its clinical goal. In particular, we wanted the monitoring result to reflect both the variation observed during treatment and the amount of treatment remaining. This led to the lead-time formulation developed in the present work, in which the projected final-course metric is accompanied by an estimate of its predictive uncertainty.

The commercial adaptive-radiotherapy software provided useful fraction-level dosimetric information, but its reporting tools did not provide the longitudinal screening and prioritization we needed. Reviewing every patient and every dose metric individually each day was not practical. Therefore, we developed an institutional dashboard that summarized ongoing treatments using information generated by the commercial software.

The dashboard allowed us to filter dose metrics according to their LT-SES probability. One useful filter identified dose metrics with probabilities between 0.50 and 0.95. In practice, limiting the initial review to dose metrics within this range reduced the number of patients requiring attention to a manageable level. Any dose metric that entered this range at any time during the treatment course was designated as a targeted dose metric and remained under ongoing evaluation for the remainder of treatment. Patients could then be ranked according to their most concerning targeted dose metrics, turning a large amount of daily information into a focused worklist.

S4.2. A Practical Choice of Dose Metrics

Our decision to use fraction-averaged scaled dose metrics (Tier 2) was largely practical (see Section 2.1 for the definition of Tier 1, Tier 2, & Tier 3 dose metrics). This approach allowed manual correction of contours generated by deformable propagation without requiring refinement of the underlying deformation map and opened the possibility of using AI autocontouring on synthetic image-of-the-day CTs. Our quality assurance of deformable dose accumulation, informed by DVH-overlay methods for evaluating deformable-registration-based dose accumulation [10], showed errors comparable in magnitude to the uncertainties associated with arithmetic summation of dose metrics and DVHs. We therefore favored a dose-of-the-day approach, using published Tier 1 clinical goals as reference thresholds while following the Tier 2 metrics longitudinally over the course of treatment.

S4.3. Choosing Alpha for Routine Monitoring

In practice, we found greater utility in choosing α according to the desired response of the monitoring signal than in using an α optimized separately for each individual time series. The purpose of α was therefore to configure how quickly the monitoring system responded to new observations, balancing sensitivity to sustained change against stability in the presence of short-term variation.

We used cohort-level one-step-ahead prediction error to help establish the lower end of the useful range for α. Across the cohorts we examined, this suggested α-values of approximately 0.20–0.30. We regarded this range as a practical lower bound for selecting α, rather than as an optimal setting for clinical use.

Values above this lower range could then be selected prospectively according to the desired signal response. This approach proved more useful operationally than time series-level optimization because it allowed the same temporal-response setting to be applied consistently across patients and dose metrics, including dose-volume metrics, while preserving α as an explicit and clinically interpretable configuration of the monitoring system.

S4.4. From Triggers to Prioritization

We initially expected that LT-SES probabilities could be converted into fixed trigger values for adaptive review. In practice, establishing useful probability triggers proved difficult. The significance of a probability depended on the dose metric, the number of treatment fractions remaining, the potential benefit of adaptation, and the resources available to act. A single probability value therefore did not naturally translate into a universal clinical action.

Prioritization, on the other hand, proved to be a more natural use of the information. Once dose metrics had been identified for ongoing evaluation, patients could be ordered according to their most concerning targeted dose metrics and reviewed from the top of the list downward. This allowed clinical attention to be directed where it was most needed without requiring a probability value to function as an automatic trigger.

Importantly, evaluation did not have to wait until a conventional trigger had been reached. Patients whose treatment trajectories appeared increasingly concerning could be queued for imaging review, contour review, dose evaluation, or possible adaptation while sufficient treatment fractions remained for an intervention to be useful.

This proactive approach fit our available resources better than either reviewing every patient intensively or waiting for a late-course trigger. Adaptive evaluation, replanning, plan review, and quality assurance require substantial clinical effort. Prioritization allowed that effort to be directed toward patients most likely to warrant closer examination and allowed those examinations to be scheduled efficiently.

S4.5. Uncertainty and the Value of a Simple Monitoring Layer

Both simple exponential smoothing and Tier 2 dose metric monitoring involve approximations. In our experience, however, these uncertainties were generally small compared with those already present in the measurement layer. The reliability of the monitored dose metric depends on daily image quality, dose-calculation accuracy, the accuracy of image registration and deformation when used, and the quality of automatic contouring. Errors or inconsistencies in these steps are carried directly into the dose metric subsequently analyzed by LT-SES.

This relative importance was not formally quantified in the present study and should therefore be understood as an observation from our clinical experience. Nevertheless, it influenced our decision to favor a relatively simple forecasting model. A more advanced statistical model cannot compensate for inaccurate or inconsistent measurements supplied to it.

As the measurement layer improves, this balance may change. Better imaging, dose calculation, automatic contouring, registration, and dose accumulation may justify the use of more advanced forecasting models. With sufficiently reliable measurements and appropriate prospective validation, predictive probabilities may eventually be calibrated to clinically meaningful actions or intervention levels.

Our initial goal was more modest. We accepted the limitations of a simple LT-SES model and Tier 2 monitoring because the dashboard still provided an additional layer of safety monitoring for patients treated in our clinic. It systematically screened ongoing treatments, identified metrics for continued attention, and helped direct limited adaptive resources toward patients who warranted closer examination. Most importantly, this could occur proactively, while there was still time to investigate the treatment and consider adaptation if appropriate.

We consider this experience an early step in developing practical, uncertainty-aware monitoring for adaptive radiotherapy. The methods described here will require further refinement and validation as measurement and adaptive workflows continue to improve. We hope this work encourages further interest in these questions within the medical physics community and stimulates additional development of practical approaches to longitudinal treatment monitoring and adaptive decision support.

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